

Management on ECMO

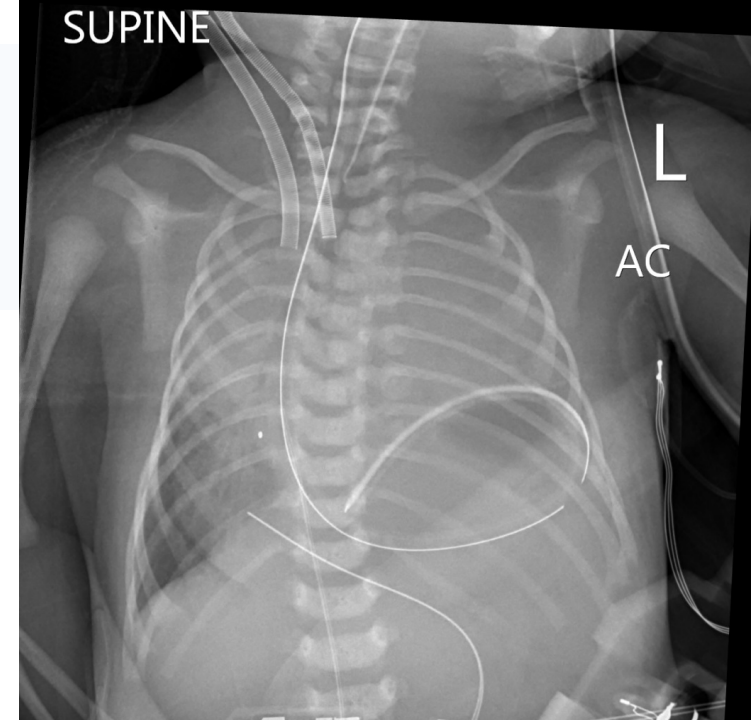
Cannulated to VA ECMO at 36h of life

When would you repair?

Planned **On-ECMO** (Super Early, Early, mid-ECMO, end of ECMO)

Salvage **On-ECMO** (repair towards end of ECMO)

Planned **Post-ECMO** (repair off)



Operative timing

When would you repair?

What are my options?

Which option gives my patient the best chance to:

1. Wean off ECLS

2. Survive

... without a devastating complication.

Planned **On-ECMO** (Super Early, Early, mid-ECMO, end of ECMO)

Salvage **On-ECMO** (repair towards end of ECMO)

Planned **Post-ECMO** (repair off)

Data

Congenital Diaphragmatic Hernia: Survival Treated With Very Delayed Surgery, Spontaneous Respiration, and No Chest Tube

By J.T. Wung, R. Sahni, S.T. Moffitt, E. Lipsitz, and C.J.H. Stolar
New York, New York

J Pediatr Surg 1995

→ resuscitation first
ok to delay surgery

Current Surgical Management of Congenital Diaphragmatic Hernia: A Report From the Congenital Diaphragmatic Hernia Study Group

By Reese H. Clark, William D. Hardin, Jr, Ronald B. Hirschl, Tom Jaksic, Kevin P. Lally, Max R. Langham, Jr,
and Jay M. Wilson, for the Congenital Diaphragmatic Hernia Study Group *J Pediatr Surg* 1998

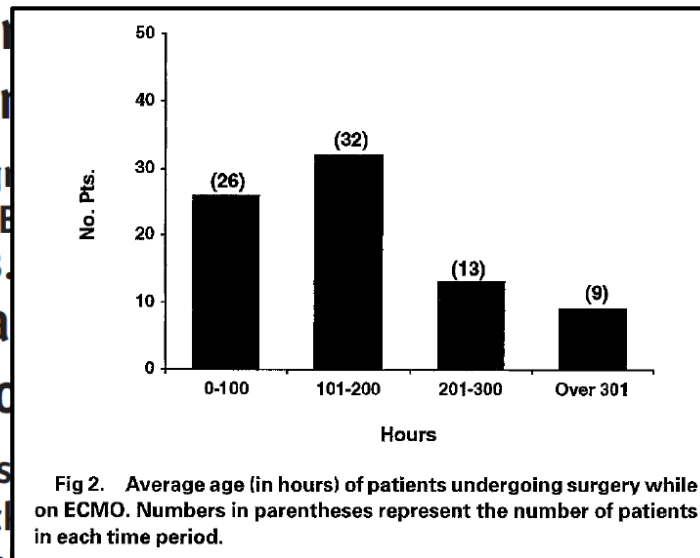
→ 57% ECLS use
33% on ECLS repair

Congenital diaphragmatic hernia membrane oxygenator

The Congenital Diaphragmatic Hernia Study Group
Brady T. West^b, Ronald B. Hirschl^a,
Pamela Lally^d, George B. Jaksic^c

Early repair extracorporeal

Melvin S. Dass^a,
Richard J. Jackson^a,
The Congenital Diaphragmatic Hernia Study Group



Extracorporeal matter? ☆

Wagner^a,
P. Lally^d,
J Pediatr Surg 2009

Extracorporeal hernia on

Wagner^b, Danny C. Little^a,

J Pediatr Surg 2010

→ introduced the controversy
after ECLS repair is better

→ wait, not so fast...
on ECLS repair is better

Data

[Surgical Repair of Congenital Diaphragmatic Hernia After **Extracorporeal** Membrane Oxygenation Cannulation **Early Repair** Improves Survival.](#)

Dao DT, Burgos CM, **Harting MT**, Lally KP, Lally PA, Nguyen HT, Wilson JM, Buchmiller TL.

Ann Surg. 2019 Aug 13. doi: 10.1097/SLA.0000000000003386. [Epub ahead of print]

PMID: 31425289

[Similar articles](#)

[Potential survival benefit with repair of congenital diaphragmatic hernia \(CDH\) after **extracorporeal** membrane oxygenation \(ECMO\) in select patients: Study by ELSO CDH Interest Group.](#)

Delaplain PT, **Harting MT**, Jancelewicz T, Zhang L, Yu PT, Di Nardo M, Chen Y, Stein JE, Ford HR, Nguyen DV, Guner Y.

J Pediatr Surg. 2019 Jun;54(6):1132-1137. doi: 10.1016/j.jpedsurg.2019.02.052. Epub 2019 Mar 1.

PMID: 30898399

[Similar articles](#)



*

Why might these data be incorrect?

Patient selection

Bias

Experience with an approach

Quality of ECLS and surgery

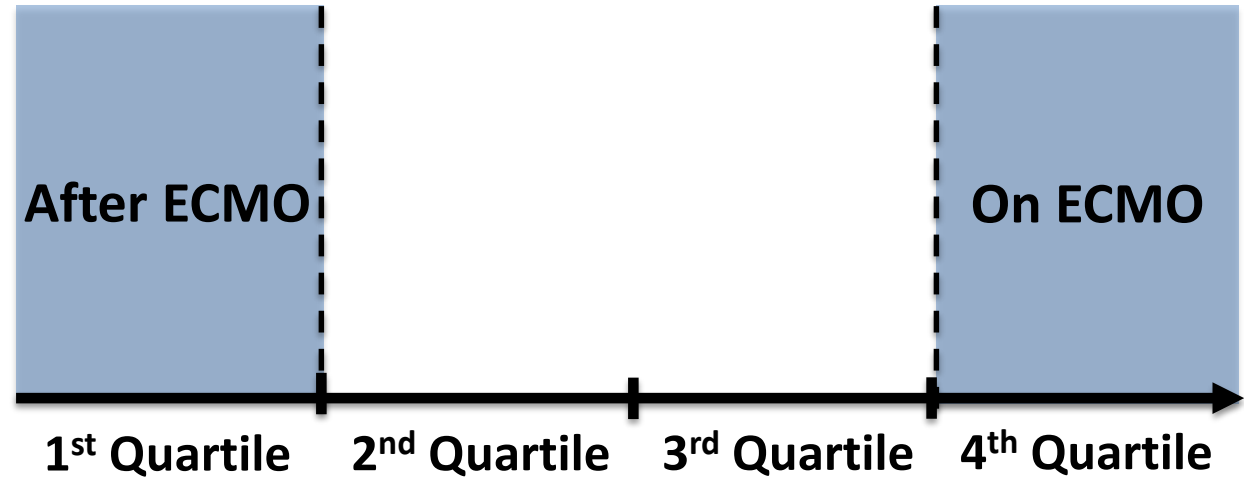
Non-repair

**Lack of understanding of factors associated with outcome
(or lack of those data)**

Attempting to address the shortcomings...

Classified by
“Intention to Treat”

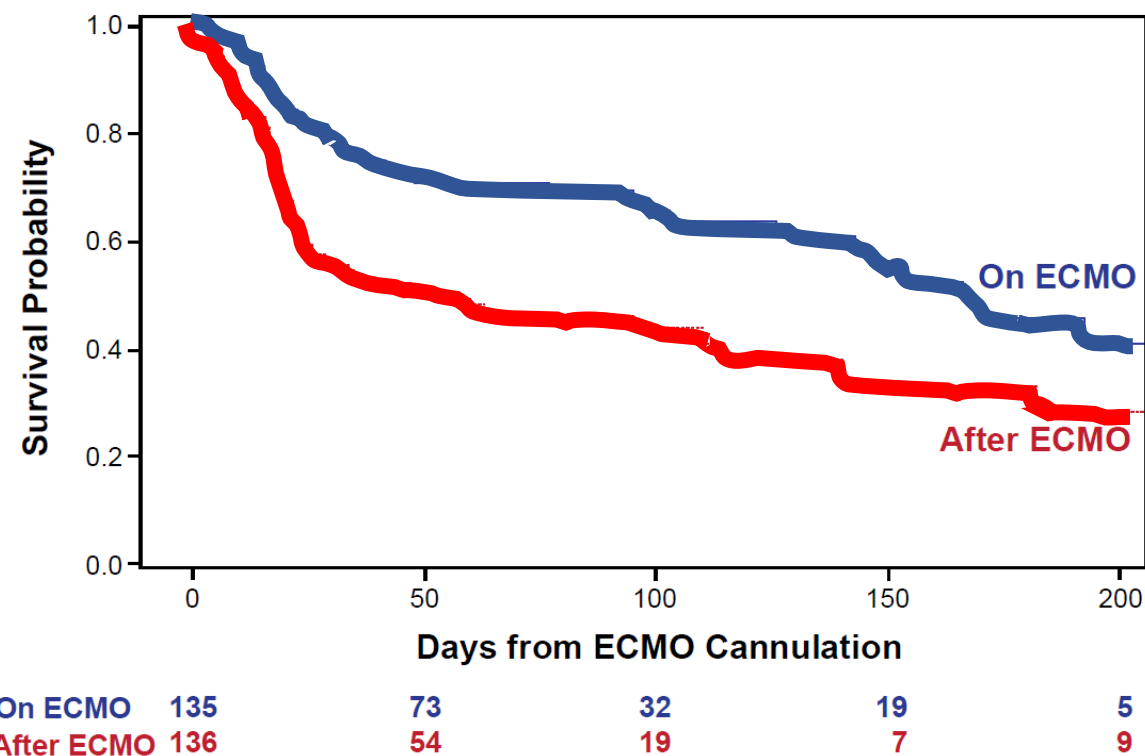
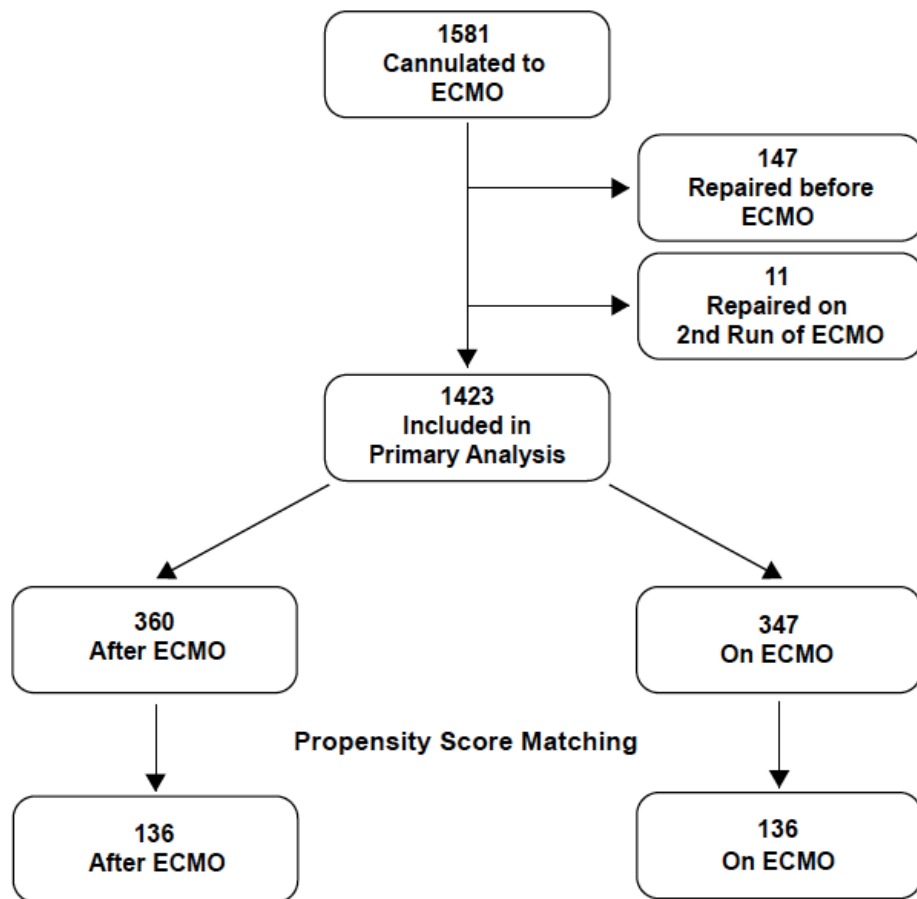
Statistical
matching



Center's Frequency of On-ECMO Repair

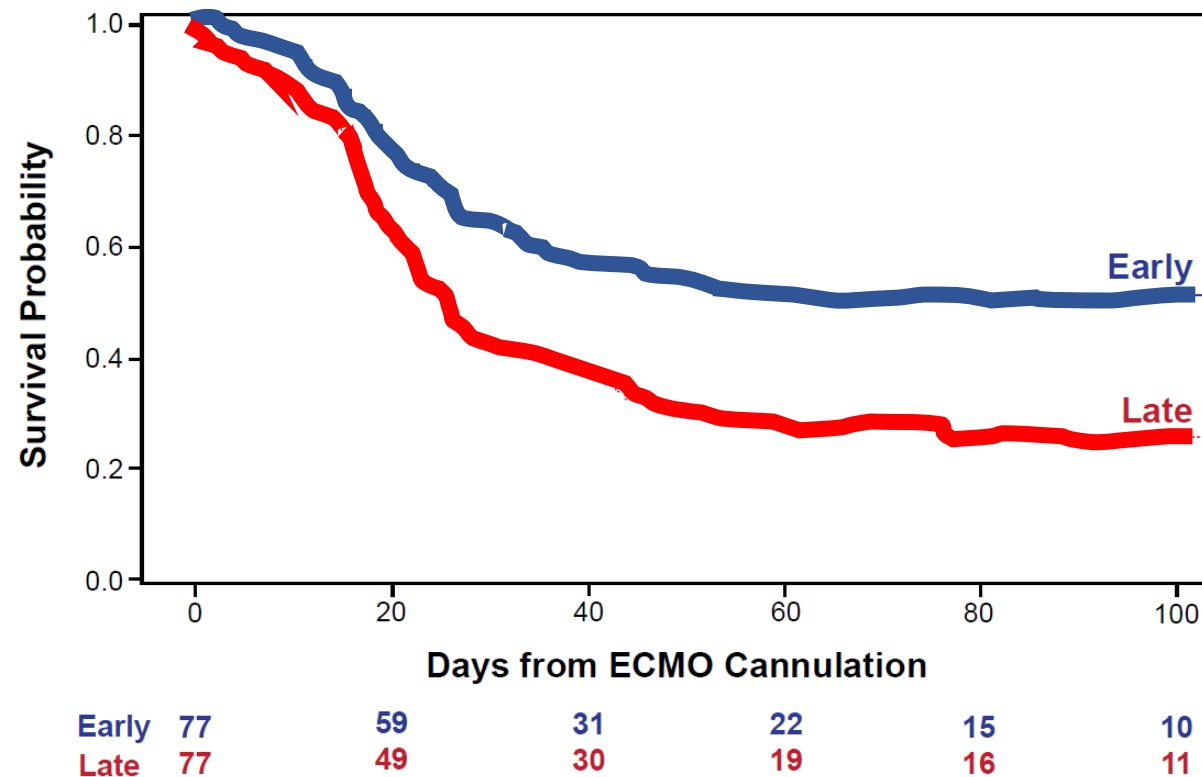
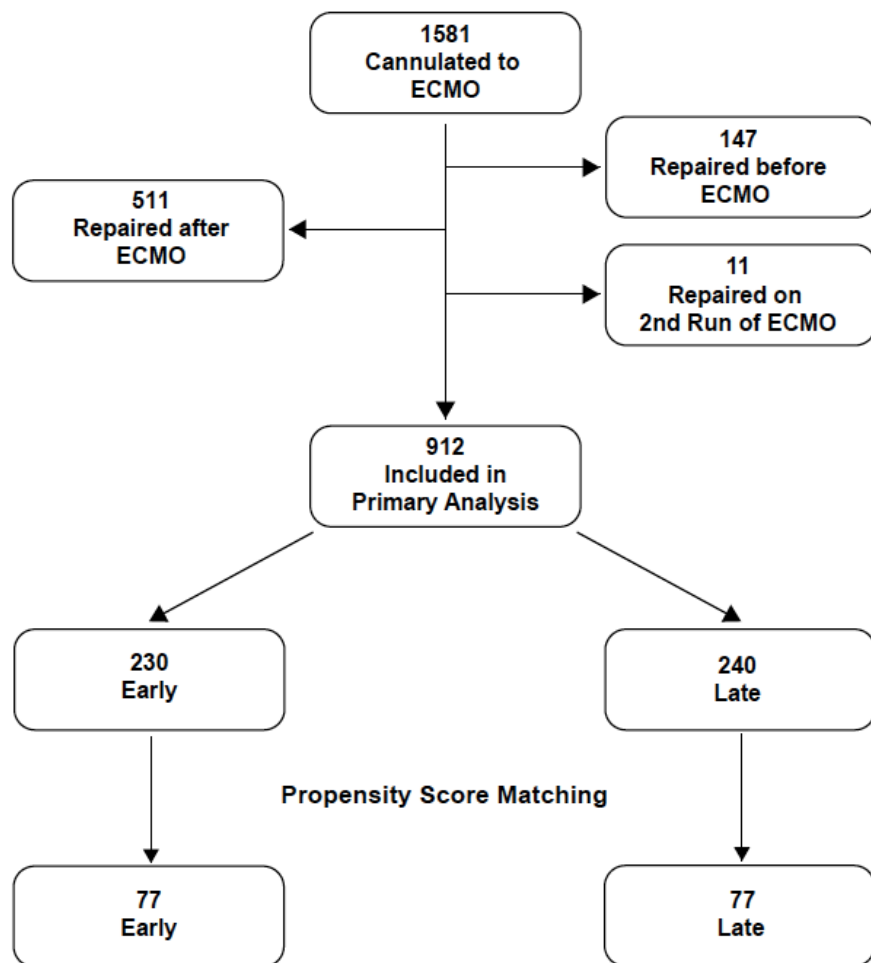
- Focuses on intention to treat
- Inclusion of non-repairs
- Propensity score matching

On ECLS vs After ECLS



Dao, et al. Surgical repair of congenital diaphragmatic hernia after extracorporeal membrane oxygenation cannulation: Early repair improves survival. *Ann Surg*, 2019

On ECLS early vs on ECLS late



Dao, et al. Surgical repair of congenital diaphragmatic hernia after extracorporeal membrane oxygenation cannulation: Early repair improves survival. *Ann Surg*, 2019

To illustrate...

On ECMO

To start with

100 patients

5 unrepaired

Repaired

95 patients

35 died (~40%)

Final survival

60 patients

After ECMO

To start with

100 patients

35 unrepaired

Repaired

65 patients

20 died (~30%)

Final survival

45 patients

Take-home message

Repair of CDH should be undertaken **early**
during the ECLS course.

Allows pulmonary expansion

Optimizes pulmonary blood flow

Eliminates the proposition of non-repair

Stabilizes cardiac compression / dysfunction

Best data *analyses* identify advantages to this approach

* Risk stratification may allow better prediction in the future